



## Detection of GBA mutations by combination of Illumina and Nanopore sequencing

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### Summary

*We are working to validate a novel Illumina caller for GBA and a pipeline based on Oxford Nanopore Technology (ONT) sequencing for GBA. Both methods are also able to detect structural variants (large reciprocal recombinant events). The Illumina caller was originally run on data obtained through the AMP-PD database. We run the ONT method on the PPMI samples included in this folder to validate both methods.*

### Method

*The ONT method has previously been reported<sup>1</sup>. It includes PCR amplification of the GBA gene, barcoding and sequencing with ONT MinION (kit SQK-LSK109).*

*To carry out this analysis, we have updated the pipeline.*

The primers used are:

-For non-recombinant alleles:

Forward 5' - TTTCTGTTGGTGCTGATATTGCTCCTAAAGTTGTCACCCATACATG -3'

Reverse 5' - ACTTGCCTGTCGCTCTATCTTCCCAACCTTTCTTCCTTCTTCTCAA -3'

-For reciprocal fusion alleles (copy number loss):

Forward 5' - TTTCTGTTGGTGCTGATATTGCGTGTCCATTCTCCATGTCTTCA-3'

Reverse 5' - ACTTGCCTGTCGCTCTATCTTCAGCCTTCCTTCCTTCCCTGCAT-3'

-For reciprocal duplication alleles (copy number gains):

Forward 5' - TTTCTGTTGGTGCTGATATTGCGTGTCCGTTCTCCACATCCTTG-3'

Reverse 5' - ACTTGCCTGTCGCTCTATCTTCCCAACCTTTCTTCCTTCTTCTCAA-3'

Bam files are aligned to GRCh38.p13.

*The pipeline produces sequencing data for the full gene, including intronic regions. We are now able to detect INDELS like 84GG.*

*Moreover, we detect reciprocal recombinant events (copy number gains and copy number losses) by means of two additional PCRs designed to detect specifically these classes of recombinants.*





*For all non-recombinant alleles, .fastq and .bam files are provided (marked as <sampleID>\_WT). When a recombinant event was detected with the additional PCRs, these amplicons were also sequenced and the resulting .fastq and .bam files are provided (marked as <sampleID>\_recombinant). A detailed methodology will be reported in a paper that is currently under submission and will be uploaded on medRxiv.org.*

## References

Leija - Salazar, M., Sedlazeck, F.J.F.J., Toffoli, M., Mullin, S., Mokretar, K., Athanasopoulou, M., Donald, A., Sharma, R., Hughes, D., Schapira, A.H.V.A.H.V.V., et al. (2019). Evaluation of the detection of GBA missense mutations and other variants using the Oxford Nanopore MinION. *Molecular Genetics & Genomic Medicine* 7, e564.

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